Q590: Order Reversal is not a single-migration growth diagram.

B-moves and C-moves commute (computed); B-moves among themselves do not

PROVE 2026-11-14 c2 — Q590 *refuted in its well-posed form*; G1 still *not proved*

Clio

2026-11-14

Provenance (PROTOCOL §2.3)

- Code: `proofs/code-1114c2-Q590/` in clio-vega/proofs, commit `64131e56a304`. That commit existed before this line was written. Engine: `code-1112c2-Q585/`, `code-1113-Q585-H1/readouts.py`, `code-1114-Q585-G1/transit.py`, all unchanged. The census generator in `common.py` is copied verbatim from `code-1114-Q585-G1/lemma3.py`: 134 elements, $c \geq 2$, $|\lambda|, |\mu| \leq 5$, $\mathfrak{gl}_4$.
- Upstream: `proofs/2026-11-14-Q585-G1-prop51.tex` @ `ee37bda9f68b` (the reduction to Order Reversal) and `proofs/2026-11-13c2-Q585-G4-evacuation.tex` (Lemma D).
- Source read first-hand this session: van Leeuwen, *The Littlewood–Richardson rule, and related combinatorics*, arXiv `math/9908099`. I read the TeX source `lrr3.tex`, which the BROWSE c2 agent had downloaded to local disk. I did no browsing in this session. See §**??**.

1 Setting

M is a mosaic with boundary (α, β, γ), $m = |\alpha|$, and $c_{\alpha\beta\gamma} \geq 2$. The readouts are $R_{\mathrm{ii}} = \mathcal{R}_A(+, N)$ and $R_{\mathrm{iii}} = \mathcal{R}_A(-, W)$. The c1 session proved the following (`thm:reduction`): $R_{\mathrm{iii}} = \iota_* R_{\mathrm{ii}}$ iff *Order Reversal* holds. Order Reversal says $c_k = \iota(b_{m+1-k})$, where b_k (resp. c_k) is the k-th box filled in B by (ii) (resp. in C by (iii)). By Cor. `cor:orders` of that file, (ii) removes the cells of α in the order R (top row first, east to west within a row), and (iii) removes them in the order D (depth 1 first, east to west within a depth). The *single migration* $x \to B$ (resp. $x \to C$) of an outer corner x of the current A-nest means one rhombus journey in direction N_A (resp. E_A). Both are the engine's `journey`. Every single migration in this session landed in the intended nest: 0 wrong-nest landings over all runs.

2 Step 0: the pre-registered control is the target

The brief asks whether the readout "reverse the (ii) B-list and apply ι" differs from (iii). That readout is the sequence $\iota(b_m), \ldots, \iota(b_1)$. Comparing it with $(c_1, \ldots, c_m)$ *is* Order Reversal, and it agrees on 134/134 (c1, `lemma3.py`). So the "control" cannot separate anything: it is the target in disguise. Agreement here is not evidence that the identity is bookkeeping, because the c1 reduction shows that it is *equivalent* to G1.

Observation 1 (not bookkeeping)**.** *(iii) is not a time reversal of (ii), at the level of journeys (`step0.py`, 606 journeys):*

- *the hexagon step-strings* $(\#a, \#b)$ *of the* k*-th (ii) journey and the* $(m{+}1{-}k)$*-th (iii) journey agree on only* 86/606*;*
- *the two journeys agree on every step in* 0/134 *cells;*
- *the total move counts differ:* 4262 *hexagon moves for (ii) against* 3612 *for (iii).*

There is also no geometric symmetry that could produce the reversal. For $d \neq n - d$ *the hexagon's symmetry group consists of the* 120° *rotations, because the side lengths and the angles both alternate. These rotations move the corner A. So no symmetry of a fixed mosaic space carries the (ii) migration to the (iii) migration.*

Verdict on Step 0. The control is the target, and the target is not bookkeeping. Order Reversal remains a genuine statement, equivalent to G1.

3 Step 1: the falsifier

The literal form is ill-posed. Q590 asks whether "the B-chain and ι(reversed C-chain) are the two boundary readings of one switching growth diagram". Both are saturated chains from β to $\gamma^\vee$. Order Reversal is the statement that they are *the same chain*. A switching diagram (Lemma D; van Leeuwen Def. 2.2.1) needs *two* input chains, the U-letters and the F-letters. Q590 names one. The only other chains carried by the data are the removal orders R and D on α. These are fixed by α and carry zero bits about M.

The well-posed form. Lemma D's mechanism is *path independence*: every monotone flip sequence produces each vertex with the same value. The analogue that the engine can build without any tableau dictionary uses ideals of α:

> (Q590′) For a fixed destination (B or C), the state after the cells of $\alpha \setminus I$ have been migrated one at a time from outer corners depends only on the ideal I and not on the order. Equivalently, every diamond commutes: $x \to X$ then $y \to X$ equals $y \to X$ then $x \to X$.

If (Q590′) held, the states would form a growth over the lattice $J(\alpha)$, and R and D would be two maximal chains in it.

Proposition 2 (Q590′ is false)**.** *On the census, (Q590′) fails for both destinations:*

destination	*ideals reached*	*ideals with* > 1 *mosaic*	*cells where it holds*
B *(dir.* N_A*)*	*1100*	*233*	*32/134*
C *(dir.* E_A*)*	*1100*	*129*	*62/134*

It also fails at the level of nest shapes*: the same 233 and 129 ideals have* > 1 *destination shape (`confluence2.py`). It fails for the* final labelled tableau*, where cells carry* f_+ *to* B *and* f_- *to* C*: 102/134 cells (B) and 72/134 cells (C) have* > 1 *outcome over all linear extensions (`finalorder.py`).*

First mismatch (`witness.py`). Take $\lambda = \mu = (2,1)$, $\nu = (3,2,1)$, the first LR filling. Then $\alpha = \beta = (2,1)$ as cell sets $\{(0,0),(1,0),(0,1)\}$, and the outer corners are $x = (0,1)$ and $y = (1,0)$.

- Order x, y lands at $(1,1)$ then $(0,2)$. The B-nest becomes $\{(0,0),(0,1),(0,2),(1,0),(1,1)\}$.
- Order y, x lands at $(1,1)$ then $(2,0)$. The B-nest becomes $\{(0,0),(0,1),(1,0),(1,1),(2,0)\}$.

Whichever corner goes first lands at $(1,1)$. The second one lands at $(0,2)$ or at $(2,0)$. Corner-independence of the landing box is not a general law either: it holds at 221/390 (B) and 110/379 (C) states with ≥ 2 corners (`sameland.py`).

Non-automatic-step statistic ([[a-generator-cap-can-skip-the-only-nontrivial-step]]). The refutation is witnessed directly, so the sample reaches the non-trivial step by construction. In the diamond census below, 440 same-direction (B,B) diamonds have journeys that *interact* (a step-string changes with the order), and *none* of them commute. **Q590 is refuted in the only well-posed form I could give it. The literal form is ill-posed.**

4 What survived: mixed commutation

Observation 3 (mixed commutation; *computed*)**.** *Let $x \neq y$ be outer corners of the current A-nest of any state reachable by single migrations. Then*

$$(x \to B)\,;\,(y \to C) \;=\; (y \to C)\,;\,(x \to B),$$

with the same final tiling and the same landing box for each rhombus.

pair type	diamonds	commute	interacting (step-string changes) / of those com
(B,C) (`mixed_general.py`)	1558	**1558**	262
(B,B), same instrument (control)	779	339	
(R_i, D_j) grid of (ii) $\times$ (iii) (`mixed.py`)	308	308	

The contrast is sharp. A B-move and a C-move commute even when their journeys interact (262 cases). Two B-moves commute *only* when they do not interact. So the order information that Order Reversal is about lives entirely in the *same-direction* interactions. The state at each vertex of the (R,D) grid $\{(i,j) : R[:i] \cap D[:j] = \emptyset\}$ is path independent, and along the grid the B-landing of $R[i]$ does not depend on j, nor the C-landing of $D[j]$ on i.

Mixed commutation does not suffice. Here is the most direct way it could have given Order Reversal. Suppose that after a full mixed emptying of A, the set L_B of B-landings depended only on $|S_B|$, the number of cells sent to B (hypothesis K). Lemma `lem:forcing` of the c1 file gives $L_C = \iota\big((\gamma^\vee/\beta) \setminus L_B\big)$ whenever A is empty. Compare "$R[:m-k] \to B$, rest $\to C$" with "$D[:k] \to C$, rest $\to B$". Under K this would yield $\{c_1,\ldots,c_k\} = \iota\{b_{m-k+1},\ldots,b_m\}$ for every k, which is Order Reversal. But K fails: 342 of 740 (cell, k) classes have > 1 landing set. Even the weaker K$'$ (L_B depends only on the set S_B) fails on 756/3376 classes (`count_hyp.py`). The two decompositions send different *sets* of cells to B. Mixed moves cannot reorder past the poset of α (the bottom cells cannot leave before the top ones), so commutation never connects the two decompositions.

Where an inductive proof would need a new lemma. Let M_1 be M after (ii) has sent the top row H of α to B, and let $\hat{\alpha} = \alpha\backslash(\text{top row})$. Under the bijection $\varphi : \hat{\alpha} \to \alpha$ that shifts the cells of the first α_ℓ columns up by one, the depth flock of $\hat{\alpha}$ matches the depth flock of α off the bottom cells of those columns. Then G1 for M follows from G1 for M_1 (induction on $\ell(\alpha)$), together with the forcing lemma, together with:

> (Exchange) the first $m - \alpha_\ell$ landings of (iii) on M coincide with the landings of (iii) on M_1.

The mixed-commutation grid gives (Exchange) for the (iii)-steps *before* D first reaches the top row (columns east of $\alpha_\ell - 1$). From that point on, (iii) on M moves the top-row cell $(c, \ell - 1)$, while (iii) on M_1 moves $(c, \ell - 2)$. These are different physical rhombi, so (Exchange) there is a genuinely new, non-commutation statement. It is not proved. I state it only as the precise location of the gap. It is not a new conjecture: on the census it is equivalent to G1 given the induction.

5 Step 3: the bypass

The mixed identity $R_{\text{iii}}(\rho^{-1}P) = \Phi(R_{\text{ii}}(P))$ was computed 134/134 on 11-13. It is Order Reversal composed with G3, and its proof needs a migration-to-jdt dictionary of Purbhoo's Prop. 5.2 type, which is also an omitted proof. G4 showed that the stepwise form of that dictionary fails (journey $k \leftrightarrow$ slide $|\mu| + 1 - k$). I made no attempt this session: the bypass moves the gap; it does not close it.

6 Coplactic lemma: locator found and read

The G4 Lemma `lem:B` (coplactic invariance) was "recalled, locator unverified". Rick (UID 915) proposed van Leeuwen. Verbatim from `lrr3.tex`, §3 "Coplactic operations", subsection "The main theorem: commutation", the first statement of that subsection (label `\copcommthm`):

> **Theorem.** Coplactic operations e_i and f_i on tableaux commute with jeu de taquin slides in the following sense. If $S \in \text{Tab}(\mu/\nu)$, $T \in \text{Tab}(\lambda/\mu)$, and $X(S,T) = (T', S')$, then $e_i(T)$ is defined if and only if $e_i(T')$ is, and if so, one has $X(S, e_i(T)) = (e_i(T'), S')$; the same holds when e_i is replaced by f_i.

(In the source, the outer shape is written with a macro for λ, which I have expanded.) The number "Theorem 3.3.1" is *inferred* from the subsection counter (section 3, third subsection, first statement). The label is generated by a macro, and I did not render the PDF. This is the switching form, and it gives Lemma `lem:B` directly: take S to be a standard filling of the inner shape. Then T' is the result of the slides, the vacated cells S' do not change, and $V(e_i T) = V(T)$. This grades the G4 coplactic lemma as **cited (read first-hand, number inferred)**. Its own proof ("the only one in our paper with a somewhat technical proof") I did not check.

7 Status and gaps

- Q590: literal form *ill-posed*; well-posed form (Q590′) *refuted* (Prop. **??**, explicit witness).
- Step 0: the control is the target. Order Reversal is not bookkeeping (Obs. **??**).
- New computed fact: mixed commutation (Obs. **??**), 1558/1558, including 262 interacting diamonds. Not proved.

- G1 / Order Reversal: still *computed* (134/134, 42/42), not proved. The gap is now located at (Exchange): the step where (iii) on M and on M_1 move different physical rhombi.
- Engine caveats inherited: the flagged tie-break interpretation of migration. Single migrations out of standard order are *outside* Purbhoo's definition, which processes whole flocks in standard order. The refutation of (Q590′) is therefore a statement about the engine's natural extension. It is not a statement about anything Purbhoo defines. Within Purbhoo's definitions, only the (R, D) grid (308/308) uses standard-order prefixes.

References

[P] K. Purbhoo, *Puzzles, tableaux and mosaics*, arXiv:0705.1184 (local seed PDF).
[vL] M. van Leeuwen, *The Littlewood–Richardson rule, and related combinatorics*, arXiv:math/9908099, §3 (TeX source, read first-hand).